IET Radar, Sonar & Navigation

Research Article

Improved method for SAR image registration based on scale invariant feature transform

ISSN 1751-8784
Received on 23rd May 2016
Accepted on 20th June 2016
E-First on 20th July 2016
doi: 10.1049/iet-rsn.2016.0261
www.ietdl.org

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Abstract: Scale invariant feature transform (SIFT) is one of the most common registration algorithms for synthetic aperture radar (SAR) images. However, the occurrence of speckle noise and geometric distortion within SAR images usually leads to limited effectiveness, challenging the stability of SIFT and its variants in real actual applications. In this study, significant improvements for SAR image registration with SIFT are made, which lie mainly in two aspects. First, a scheme is developed to enhance the description of keypoints with improved dominant orientation assignment and support region. Second, an optimised matching method for further enhancing the matching performance is developed to reduce the mutual interference among the keypoints with similar location and dominant orientations. Extensive experiments confirm the effectiveness of the proposed algorithm for SAR images.

1 Introduction

Image registration is a key step for synthetic aperture radar (SAR) applications, such as navigation, change detection, and hitting effect evaluation [1–3]. In SAR image registration, feature-based algorithms present great advantages over those based on intensity or area feature [4–6]. Among them, scale invariant feature transform (SIFT) [7] is one of the most widely applicable algorithms owing to its high effectiveness in describing local features of SAR images.

However, in application of SAR image registration, some important characteristics of SIFT are degenerated, due to the fact that SIFT is easily influenced by the speckle noise and geometric distortion, which results from the scattering structure of radar with more complex mechanism. In current literature, many modifications of SIFT have been developed to improve its performance in SAR image registration. Wang *et al.* [8] adopted a novel bilateral filter SIFT (BF-SIFT) method with the scale space built by a bilateral filter. It enhanced the applicability to SAR images with different aspects and wavebands. Wang *et al.* [9] proposed the adapted anisotropic Gaussian SIFT (AAG-SIFT) matching strategy for SAR image registration by applying an adapted anisotropic Gaussian filter, which was proved to be robust to noise and able to retain more details. Dellinger *et al.* [10] proposed a SIFT-like algorithm for SAR image registration, namely SAR-SIFT. In this algorithm, a multi-scale SAR-Harris matrix based on the multi-scale Harris detector was used to offer stable keypoints in SAR images, and thus robust gradient orientations calculation by gradient by ratio was applied to obtain more distinctive feature descriptor. Wang *et al.* [11] further improved the SAR-SIFT algorithm with uniformly distributed features on the basis of a Voronoi diagram to select stable keypoints and extract optimal features. In the aforementioned algorithms, the improvements lie mainly in two aspects. First, more adaptive scale spaces were put into use to eliminate the instability of Gaussian scale space existing in the original SIFT algorithm. Second, more robust features were extracted with the reliable orientation and description for keypoints. Although the above extensions of SIFT algorithm were improved considerably with more robust and reliable performance, there is still some room for improvement. Due to the interference of speckle noise and the low-contrast characteristics of SAR images, it was still unstable in the dominant orientations (DOs) assignment. As a result, lots of keypoints were invalidly described in the SAR image registration. The phenomenon was more serious when the SAR image pairs are imaged with an angle change. Though Fan *et al.* [12] and Suri *et al.* [13] suggested suppressing the orientation assignment for accuracy and efficiency, the improvement is limited. Hence, the correct matching rate is very low in SAR image registration. On the other hand, the mutual interference among keypoints reduces the matching performance with the current matching methods.

In this paper, an enhanced SAR image registration strategy on the basis of SIFT is introduced with two improvements. First, an improved orientation assignment and descriptor calculation to achieve robust feature descriptors are adopted for SAR images. Second, the matching method for eliminating the mutual interference among the keypoints with similar location and DOs is optimised by calculating the spatial relationship. Experiments demonstrate that the proposed strategy can reduce the interference of speckle noise and geometry distortions for SAR image matching. The main contributions of this paper can be summarised as follows.

i. This work can increase the correct matches with the same quantities of keypoints and has the invariance to scale and rotation, which is meaningful for more matches required especially in the area with fewer keypoints detected.
ii. This work can eliminate the mutual interference among the keypoints with similar location and DOs in the matching step, which further reduces leakage matches.
iii. The proposed methods can be seamlessly embedded into the current SIFT-like registration algorithms, ensuring a more generalised performance.

2 Effective feature extraction and eliminating interference nearest neighbour distance ratio (EI-NNDR) matching method

Conventional SIFT algorithm usually consists of the following core stages. The scale space is built to identify the extrema as potential keypoints. At each candidate location, a detailed model is fit to determine location and scale, and then the keypoints are selected on the measures of their stability. Afterwards, a series of operations are done on the keypoints. First, one or more orientations are assigned. Then one single scale-dependent support region around each keypoint is used for descriptor construction. The stability of DOs determines the robustness of feature description, which affects

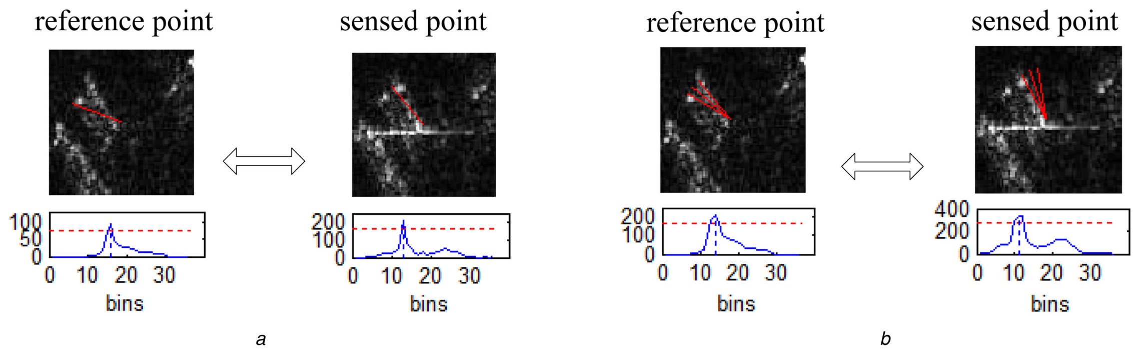


Fig. 1 *DOs of keypoints in reference image and sensed image*
(a) DO selected by SIFT method, ***(b)*** DOs selected by the new method

the matching results. The correct matching rate is low by the SIFT algorithm. Only a few detected keypoints can obtain correct matching points. An important reason is that the SIFT descriptors are constructed with the unstable DOs. In conventional SIFT registration, 36 bins of histogram covering the 360° range of orientations are calculated with a local histogram of gradient orientations, which are weighted by the gradient magnitudes and a Gaussian window. All the peaks larger than 80% of the highest peak are selected as the DOs. On the basis of the SIFT DO assignment method, the DOs in different SAR images cannot be assigned consistently, because they are affected by speckle noise and distortion. The robustness and accuracy of the DOs assignment decrease dramatically, resulting in unsuccessful SAR image registration. Another reason for the low correct matching ratio is that the texture structure is provided with low contrasts in SAR images. The single support regions contain incomplete information for keypoint description. Besides, the adjacent orientations belonging to the same keypoint also limit the increase of correct matches. Consequently, the ratio of incorrect matched keypoints to all detected keypoints stays high in SAR image registration.

In this paper, a new manner is proposed to enhance the correct matching ratio of the keypoints for SAR image registration. First, a more robust and reliable SIFT descriptor is constructed by a new assignment method with enlarged scope of bin and a larger support region with accurate texture structure. Second, a stable matching method is introduced by eliminating the mutual interferences among keypoints.

2.1 Effective feature extraction

2.1.1 Dominant orientation selection: Point spread function (PSF) estimation is a key issue influencing the quality of SAR imaging. PSF has a cross shape. Though the window function is joined to suppress it, the sidelobes still exist. The target will be interfered with the sidelobes from an adjacent strong echo target. At the same time, there is a certain relationship between the directions of the PSF cross and the viewing angle of SAR. The sidelobes influences of SAR images with different viewing angles are not consistent. Therefore, different from optical images, the pixels in SAR images are more closely correlated with each other. Thus, the DOs should be calculated with more adjacent pixels to enhance accuracy. In this work, considering the invariance to rotation, 36 orientation histogram bins covering the 360° range of orientations are formed from the gradient orientations of sample points within the support region centred at the keypoints. The scope of each histogram bin is made larger than that in optical images. A number of experiments with SAR images including the scenes of forest, farmland and city proper conclude that the bin with 30° achieves good registration performance. When the scope of the bin is bigger than 30°, the increase of degree cannot provide more correct matches. Therefore, the histogram bin with 30° is used in this paper. The DOs are selected by calculating the maxima and the values above 80% of them. Compared with the SIFT method in Fig. 1*a*, the DOs of the keypoints with the new method in Fig. 1*b* have better robustness. It is usually the case that a keypoint is corresponding to several feature descriptions when the keypoint is assigned multiple orientations. Thus, the directional keypoints are presented as $P_i = \{(x_i, y_i, \theta_i, \sigma_i), i = 1, \dots, M\}$, where (x_i, y_i) is the coordinate location, θ_i is the orientation, σ_i is the scale and K is the total number of keypoints with different orientations.

2.1.2 More seed points calculation: In this work, descriptors consider more texture structure around the keypoints, which contain more background information of the keypoints. For SAR images with different view angles, grey distribution around keypoint changes greatly, resulting in unstable SIFT descriptors and false matching. However, the big texture structure, stable and robust, can describe the local region more validly. Consequently, we adopt 5×5 seed points instead of 4×4 with the same distance between neighbouring seed point to describe the features. A set of descriptors $F = (x_1, x_2, \dots, x_M)^{\mathrm{T}} \in R^{M \times D}$ in D-dimensional space are achieved, where D is the length of one feature vector, and M is the number of the feature descriptors with independent orientation.

2.2 EI-NNDR matching strategy

NNDR and dual matching (DM) are the widely used matching strategies along with SIFT [8, 14, 15]. In the NNDR method, the best candidate match for each keypoint is found according to the ratio of the nearest neighbour (NN) distance to the second NN distance. First, a set of features from the reference image and sensed image are computed, which is denoted as $F = \{(F_i^1, F_j^2), i = 1, \dots, M; j = 1, \dots, N\}$, and the corresponding feature points are $P = \{(P_i^1, P_j^2), i = 1, \dots, M; j = 1, \dots, N\}$, where M is the number of feature descriptors with independent orientation in the reference image and N is that in the sensed image. Second, the Euclidean distances of one feature in the reference image and all the features in the sensed image are calculated and sorted in ascending order which are denoted as $D_p (p = 1, \dots, N)$, and the order of the corresponding points in the sensed image also changes and is denoted as $\tilde{P}_p^2 (p = 1, \dots, N)$. Finally, the ratio of distances is obtained by

$$\text{ratio} = \frac{D_1}{D_2} \tag{1}$$

where D_1 and D_2 are the NN distance and the second NN distance, respectively. Specifically, when ratio < thresh, it means that the matching is correct; otherwise the matching is incorrect, and thresh is set according to experiences. The DM strategy is based on the NNDR matching strategy, but double check is adopted. Actually, some keypoints are assigned with multiple orientations, and the feature vector is constructed with independent orientation. When there is little difference between the orientations of the same keypoint, the feature vectors are also similar, which means that D_2 and D_1 may have similar value. In this case, the superfluous orientation will lead to false matches in NNDR and DM strategies. On the other hand, the matching accuracy is lower than that in optical images concluded by using BF-SIFT and AGG-SIFT methods [8, 9], especially on the SAR images from different viewing angles. Therefore, the spatial relationships among the

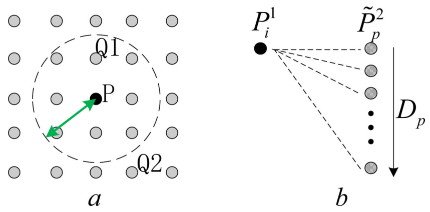


Fig. 2 *Point relationships in image*
(a) Relationships of points in one image, ***(b)*** Relationships of the point in the reference image and the potential matching points in the sensed image

matching points are an important characteristic for improving the matching performance, which is ignored in the NNDR and DM strategies. To overcome this drawback, an EI-NNDR algorithm is proposed

$$\mathrm{dis}(P,Q) = \left[(x_1 - x_2)^2 + (y_1 - y_2)^2\right]^{(1/2)} \tag{2}$$

First, the relationship of the points is defined as follows: $P(x_1, y_1)$ and $Q(x_2, y_2)$ represents two points. Then, the distance d between the two points is calculated according to (2). The matching accuracy value prmse is estimated according to experiences. When $d <$ prmse, Q is the neighbour point of P, the $Q1$ point in Fig. 2*a*, for example. When $d \geq$ prmse, Q is out of the range of point P, the $Q2$ point in Fig. 2*a*, for example. Especially, when $d = 0$, it means Q and P are the same point.

The D_p between reference point P_i^1 and corresponding points $\tilde{P}_p^2 (p = 1, 2, \ldots, N)$ is shown in Fig. 2*b*. When the point $\tilde{P}_2^2$ is the neighbour point of $\tilde{P}_1^2$ with similar DO, the descriptors will be quite similar. According to the NNDR matching strategy, the matching point $\tilde{P}_1^2$ and the reference point P_i^1 are not matching pairs. Therefore, to avoid leakage matching, it is necessary to eliminate the mutual interference of the second NN. Then the points $\tilde{P}_p^2$ are renewed and the ratio is further calculated to obtain correct matches. The procedure of EI-NNDR is outlined in Fig. 3, and the successful probability of matching is greatly increased by the EI-NNDR method.

3 Experiments

To verify the effectiveness of the proposed algorithm for SAR images, three groups of experiments are carried out. We first investigate its transferability by embedding the proposed methods into the exiting improved SIFT-like methods. Then, we verify its advantages in the invariance to scale and rotation inherited from SIFT-like algorithms. Finally, since SAR images are accompanied by much strong speckle, experiments on simulated images with speckle noise are carried out to assess the ability of the proposed algorithm to deal with speckle noise.

The framework of our experimental methods is as follows. First, the improved effective features (see Section 2.1) are extracted from the SAR images. Then, a set of matched points are obtained by dual EI-NNDR matching method, which is also noted as eliminating interference DM (EI-DM) (see Section 2.2). In EI-DM matching method, we set the parameter prmse to be 1 in the eliminating interference algorithm. That is to say, the interference of the features constructed from the same keypoint with NN distance is eliminated. In order to show the effectiveness of each modification of SIFT, four methods are evaluated.

i. **-SIFT*: SIFT algorithm with DM matching method.
ii. **-SIFT-EI*: Refers to the SIFT method by applying the modification to the matching method with EI-DM matching strategy in Section 2.2.
iii. **-SIFT-EIM1*: Refers to feature exaction in the SIFT method by applying the first modification in Section 2.1.1, and with EI-DM matching method in Section 2.2.

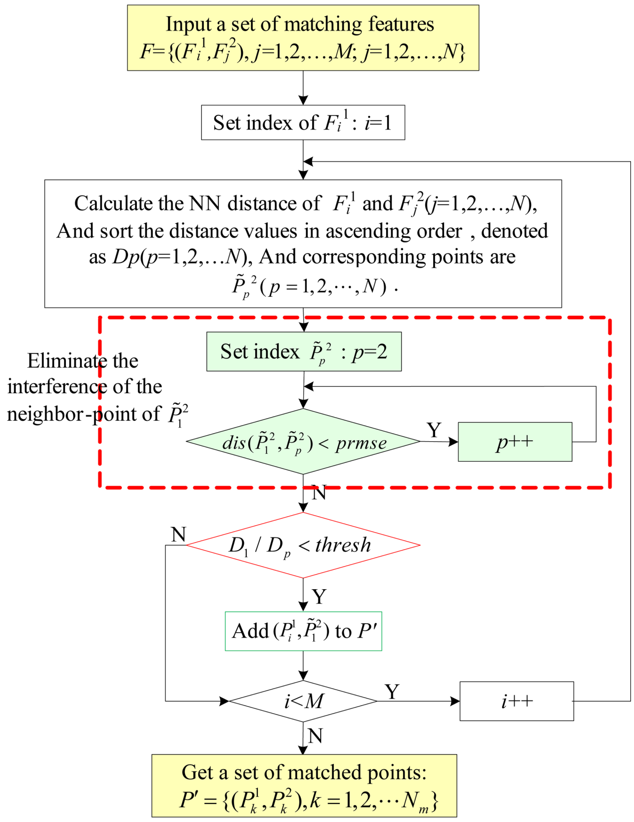


Fig. 3 *EI-NNDR algorithm*

iv. **-SIFT-EIM2*: Refers to feature exaction in the SIFT method by applying the modification in Sections 2.1.1 and 2.1.2, and with EI-DM matching method in Section 2.2.

In this paper, the proposed algorithm is evaluated with the help of precision–recall graphs [15]. The criteria are calculated as follows:

$$\text{recall} = \frac{\#\,\text{correct matches}}{\#\,\text{correspondences}}, \quad 1 - \text{precision} = \frac{\#\,\text{false matches}}{\#\,\text{all matches}} \tag{3}$$

In (3), the precision–recall graphs are based on the quantities of correct matches and false matches obtained for an image pair. The quantities of #all matches are calculated using the DM and EI-DM methods. In the set of matched points $P' = \{(P_k^1, P_k^2), k = 1, 2, \ldots, N_m\}$, the matches are defined as #all matches. Let $M(P_i^1, P_j^2)$ be a match between a point $P_i^1 = (x_i, y_i, \sigma_i, \theta_i)$ of reference image and a point $P_j^2 = (x_j, y_j, \sigma_j, \theta_j)$ of sensed image, where (x, y) is the position, σ is the scale and θ is the orientation of the keypoint. Considering the deformation A of the reference image in comparison with sensed image, $M(P_i^1, P_j^2)$ is defined as correct if

$$\| A(x_i, y_i) - (x_j, y_j) \| < a \cdot \min(\sigma_1, \sigma_2) \tag{4}$$

Here, the parameter a is empirically set up to 5; σ_1 and σ_2 are the scale of the reference image and sensed image. Then the #false matches are equal to the difference between #all matches and #correct matches.

Besides, the number of correct matches (NOM) and root-mean-square error (RMSE) [16] are used as criteria to carry out the evaluation for automatic registration methods. For the set of matched points $P' = \{(P_k^1, P_k^2), k = 1, 2, \ldots, N_m\}$, random sample consensus (RANSAC) method [17] is used to find out a set of inliers. RANSAC is an iterative method to estimate parameters of a mathematical model from a set of observed data which contains outliers. We choose $k = 10$ as the number of the fixed iterations and

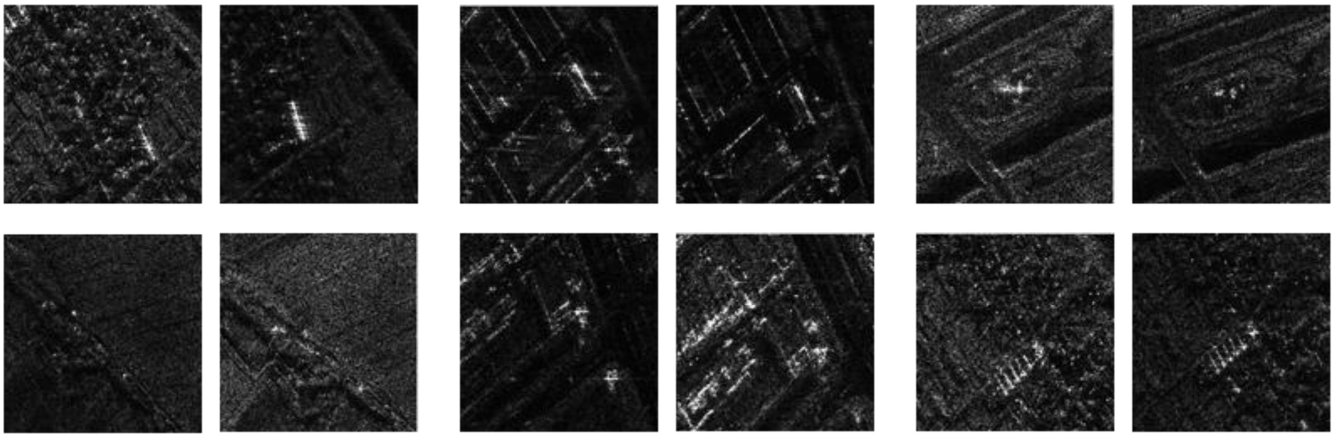


Fig. 4 *Examples for SAR image pairs with different view angles from a variety of scenes*

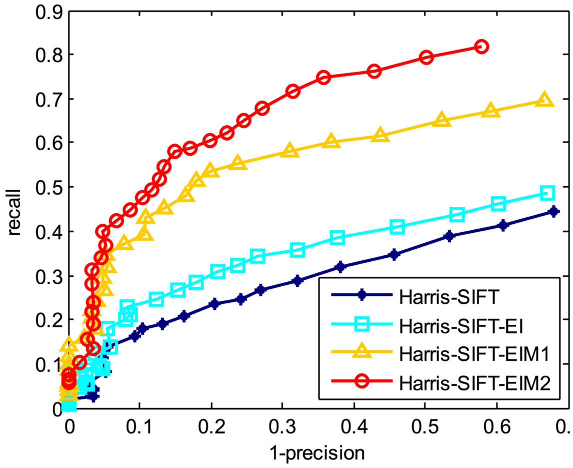


Fig. 5 *Recall versus 1−precision graphs for the four methods computed on 15 image pairs. The graphs depict the overall results for detection, description and matching steps jointly for each of the modification*

affine transformation as the global deformation. Then the set of inliers (P_c^1, P_c^2) and deformation H are used to produce the registered images. In this paper, let the number of inliers be NOM.

RMSE is computed to judge the alignment accuracy by

$$\mathrm{RMSE} = \left(\sum_{c=1}^{N_{om}} \frac{1}{N_{om}} \left(\| H(P_c^1, P_c^2) \|^2 \right) \right)^{(1/2)} \quad (5)$$

where P_c^1 and P_c^2 are the inliers, which are the matching points in reference image and sensed image. The numbers of them are both N_{om}.

3.1 Experiments to assess the transferability based on Harris-SIFT and BF-SIFT algorithms

In this part, 15 real airborne X-band SAR image pairs are used for evaluation. These images were acquired over Shaanxi, China with a resolution of 0.5×0.5 m and multiple viewing angles. The experimental images are varied. They are imaged in various scenes including forest, farmland, city proper and so on (see Fig. 4).

First, four methods of Harris-SIFT, Harris-SIFT-EI, Harris-SIFT-EIM1 and Harris-SIFT-EIM2 are evaluated based on Harris-SIFT method [18] with LoG-Harris detector and circular-SIFT descriptor.

Fig. 5 depicts the recall versus 1−precision graphs based on Harris-SIFT. As can be seen, a superior result is achieved by Harris-SIFT-EIM2. Indeed, for a 1−precision rate of 10%, almost 40% of the correct matches are obtained, and the recall rates sometimes are 40% higher than Harris-SIFT. The performances of Harris-SIFT-EIM1 and Harris-SIFT-EI methods are also better than that of Harris-SIFT with the same number of keypoints.

An example for the automatic registration when the distance ratio equals 0.8 is manifested in Figs. 6*a* and *b*. The NOMs of the Harris-SIFT method are indicative of its arduousness for SAR image registration. The proposed algorithm with EI-DM performs well in increasing NOMs as well as stable average RMSEs and time for all kinds of SAR images. The test methods with the new orientation assignment and more seed points also increase the NOMs greatly, which indicate the proposed algorithm is effective for SAR images based on Harris-SIFT.

In order for the modifications in this work to be seamlessly embedded into other existing improved SIFT-like methods, the experiments based on BF-SIFT algorithm are also evaluated. A bilateral filter [19] (with $\sigma_s = 2$ and $\sigma_r = 0.2$) is used in this method, and the following methods, BF-SIFT, BF-SIFT-EI, BF-SIFT-EIM1 and BF-SIFT-EIM2, were evaluated. An example for the automatic registration when the distance ratio equals 0.8 is manifested in Figs. 6*c* and *d*. The NOMs of the BF-SIFT matching methods indicate that using the bilateral filter instead of Gaussian filter, NOMs in the scenes including more targets can be increased. However, in the scenes including fewer targets, such as farmland, the BF-SIFT method offers a limited enhancement compared with Harris-SIFT, as shown in Fig. 6*c*. BF-SIFT-EI, BF-SIFT-EIM1 and BF-SIFT-EIM2, which are based on BF-SIFT, can achieve effective matching results. The NOMs of the four methods demonstrate that the proposed improvements in this paper can increase greatly in various scenes. The RMSEs of the four matching methods are manifested in Fig. 6*d*. Among the 15 pairs of images, the RMSEs results of the proposed methods in this paper are comparable with BF-SIFT except for images 8# and images 10#, which have fewer correct matches and result in unsafe RMSEs.

The proposed methods in this work can also be embedded into other SIFT-like registration algorithms such as AAG-SIFT and SAR-SIFT. In AAG-SIFT [8], an adapted anisotropic Gaussian filter is used to retain more details in the description, and it is robust to noise. The DO consistency matching method in AAG-SIFT can remove outliers, but the initial matching set was obtained by the NNDR method. In SAR-SIFT [10], the improvements are done for stable keypoints and robust gradient orientations calculation, and the orientations assignment adopts conditional SIFT methods. Furthermore, the matching method of SAR-SIFT is also based on NNDR method. Therefore, the proposed orientation assignment method and the EI-NNDR or EI-DM method can be used in AAG-SIFT and SAR-SIFT and achieve better performance.

3.2 Experiments to assess the invariance to scale and rotation

The invariance to scale and rotation is the advantages of the SIFT-like algorithms, so we carry out a group of experiments to evaluate the invariant of the proposed methods. In this part, 18 real airborne X-band SAR image pairs with a resolution of 3×3 m are used for evaluation. The scale ratio of the sensed images and reference images is 0.625. In addition, the sensed images are rotated with 90° in clockwise direction.

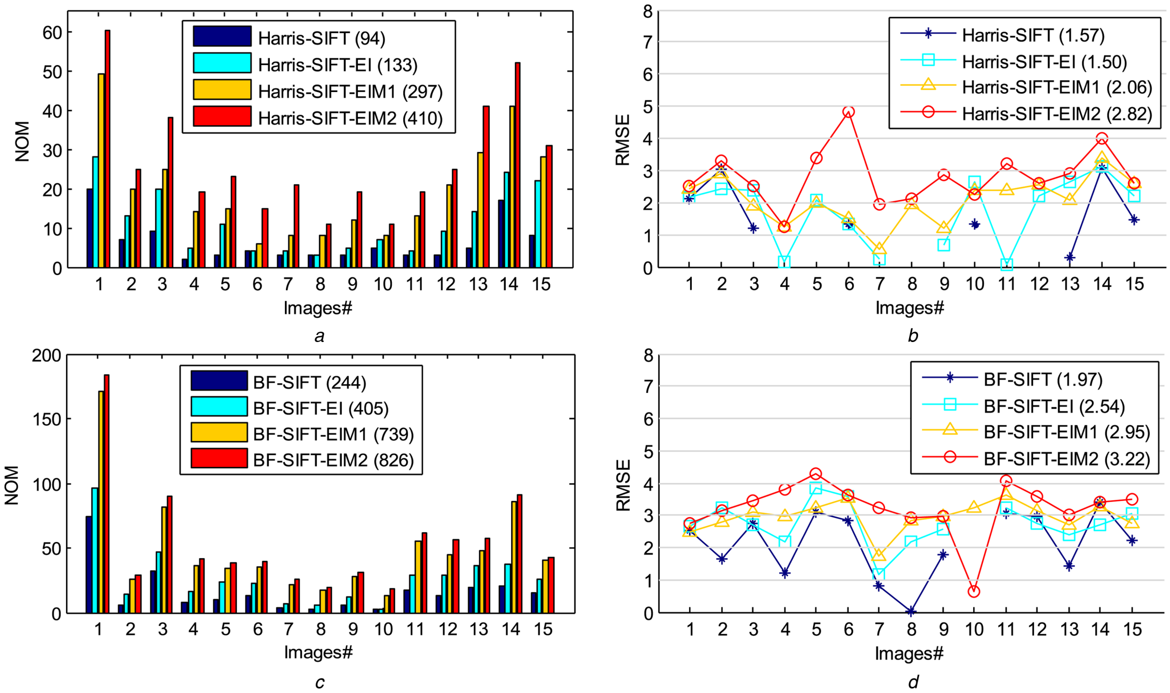


Fig. 6 *Experimental results on SAR images when the distance ratio equals to 0.8 in matching step*
(a) NOM for 15 SAR image pairs based on Harris-SIFT. In parenthesis, next to the name of each of the methods, we show the sum of NOM of all image pairs, ***(b)*** RMSE for 15 SAR image pairs based on Harris-SIFT. In parenthesis, next to the name of each of the methods, we show the average RMSE of all image pairs, ***(c)*** NOM for 15 SAR image pairs based on BF-SIFT. In parenthesis, next to the name of each of the methods, we show the sum of NOM of all image pairs, ***(d)*** RMSE for 15 SAR image pairs based on BF-SIFT. In parenthesis, next to the name of each of the methods, we show the average RMSE of all image pairs

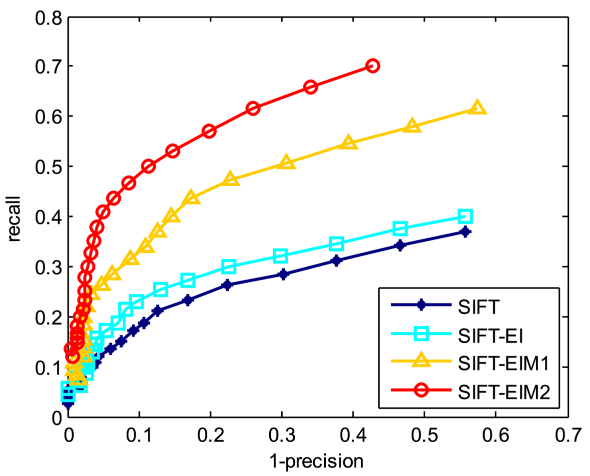


Fig. 7 *Recall versus 1−precision graphs computed on 18 image pairs with different scales and different orientations. The methods detect keypoints with the same method*

Fig. 7 depicts the recall versus 1−precision graphs computed on 18 image pairs with different scales and orientations. We find that SIFT-EIM2 achieves better performance, and the recall rates sometimes are 40% higher than SIFT. SIFT-EIM1 and SIFT-EI methods also perform better than that of SIFT with the same number of keypoints. The graphs indicate clear that the proposed methods inherit the invariant to scale and rotation and perform better. Here, we also take an example of automatic registration and set distance ratio to 0.8. The NOM and RMSE of the SAR image pairs are shown in Fig. 8.

From Fig. 8*a*, we find that the proposed algorithm can obtain a larger number of matches for SAR images with different scales and orientations in various scenes. Among the 18 pairs of images, the four methods in Fig. 8*b* also have comparatively high matching precision. The values of RMSE of SIFT method are slightly smaller because there are fewer correct matching numbers, which leads to unsafe RMSE. Figs. 7 and 8 indicate that the proposed methods inherit the invariance to scale and rotation from SIFT-like algorithms and achieve better performance. Fig. 9 gives two registration results of image #3 and image #17.

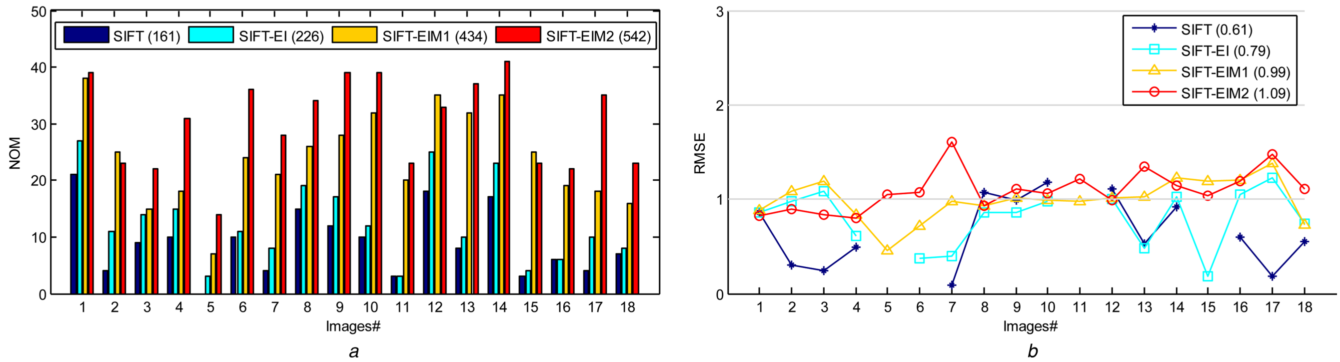
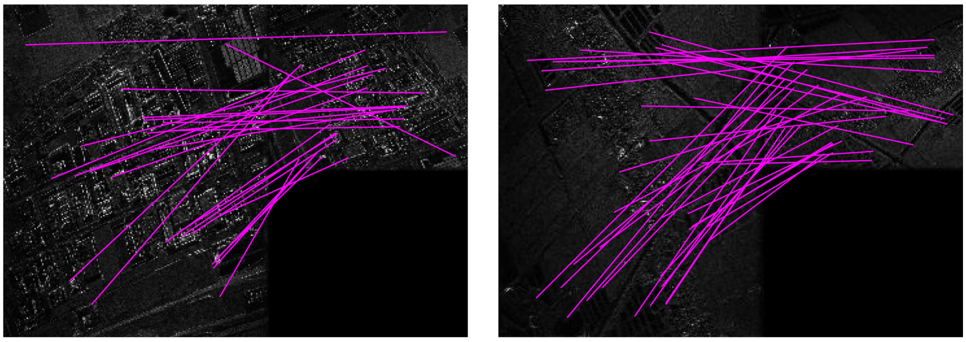
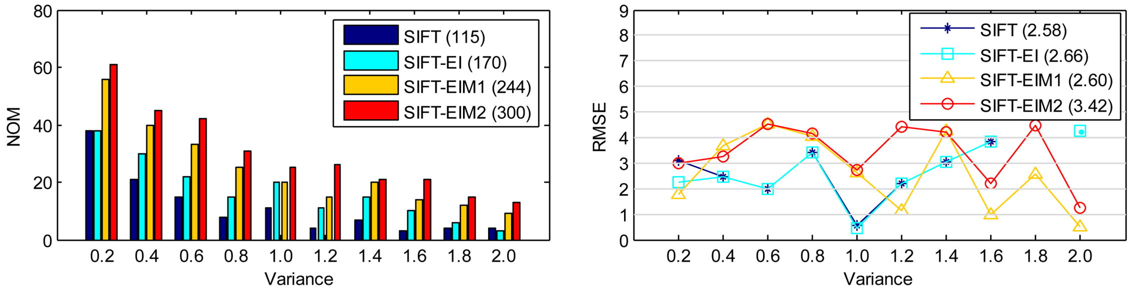


Fig. 8 *Experimental results on SAR images with different scale and different orientation when the distance ratio equals 0.8 in matching step*
(a) NOM for 18 SAR image pairs. In parenthesis, next to the name of each of the methods, we show the sum of NOM of all image pairs, ***(b)*** RMSE for 18 SAR image pairs. In parenthesis, next to the name of each of the methods, we show the average RMSE of all image pairs



a b

Fig. 9 *Registration results on SAR images with SIFT-EIM2 method; the scale ratio of the sensed images and reference images is 0.625*
(a) SAR image pair #3 with the scene of city proper, ***(b)*** SAR image pair #17 with the scene of farmland



a b

Fig. 10 *Experimental results on simulated images with speckle noise when the distance ratio equals 0.8 in matching step*
(a) NOM for ten image pairs. In parenthesis, next to the name of each of the methods, we show the sum of NOM of all image pairs, ***(b)*** RMSE for ten image pairs. In parenthesis, next to the name of each of the methods, we show the average RMSE of all image pairs

Table 1 Average running time (second)

Method	SIFT	SIFT-EI	SIFT-EIM1	SIFT-EIM2
Harris-SIFT for SAR images	5.19	5.09	8.78	10.07
SIFT for simulated images	5.31	5.41	8.24	9.57

3.3 Experiments to assess the ability to deal with speckle noise

Speckle noise widely exists in the SAR images, which lead to low matching performance [20]. To verify the validity of the proposed methods for images with speckle noise, SIFT, SIFT-EI, SIFT-EIM1 and SIFT-EIM2 are evaluated with the same keypoints. The four methods are carried out on simulated images with different speckles. We add multiplicative noise to the Lena image I, using the following equation:

$$J = I + n \times I \tag{6}$$

where n is uniformly distributed random noise with mean $m = 0$ and variance $v = 0.2, 0.4, \ldots, 2$. The four methods are used to find correct matching points between the original image I and the noised image J. The experimental results are shown in Fig. 10.

The NOMs of the four matching methods decrease rapidly with strong speckle. This indicates that it is difficult to register the SAR images with strong speckle. The proposed improvements perform better in reducing the influence of speckle noise. They can always find more correct matching points than SIFT methods. There is little difference among the RMSEs of the four matching methods, which indicates the proposed improvements have stable matching accuracy. Note that compared with DM matching methods, the proposed EI-DM matching strategy can achieve great increase in NOMs for SAR images. The difference between the values of RMSEs and the average running time and that of the SIFT algorithm is subtle, which indicates the EI-DM algorithm is robust both in matching accuracy and in matching efficiency. SIFT-EIM1 has the approximate RMSEs, but the RMSEs in SIFT-EIM2 method is slightly larger but still within the acceptable scope.

As shown in Table 1, compared with that by SIFT and SIFT-EI methods for SAR images and simulated images, the running time of SIFT-EIM1 and SIFT-EIM2 is increased, which is due mainly to the effective feature extraction with increased feature description and larger support region calculated by more seed points. However, they are meaningful for more matches required especially in the scenes with difficult keypoints detection.

4 Conclusions

This paper presents some improvements for SAR image registration based on the SIFT-like algorithms. The matching ability of the proposed algorithm to resist speckle noise and distortion for SAR image registration is promoted from two aspects. A more stable DOs assignment method is adopted to increase the orientation consistency; the description with larger support region is calculated to obtain more local information. Besides, the matching strategy to eliminate the mutual interference among keypoints in NNDR or DM method maximum increases the NOMs without the influence of RMSEs. As manifested in the experiment section, the improvements of the proposed algorithm can be seamlessly embedded into current registration algorithms with the evaluation of matching accuracy and efficiency.

5 Acknowledgment

The authors thank all of the anonymous reviewers for their valuable comments and suggestions to improve the paper's quality. This work was supported by the China National Natural Science Foundation under the grant 61401363.

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